VIDEO SCRIPT

FAIR LOAD-SHEDDING WITH QAOA: 3-MINUTE PITCH

Team Winners: Usman Saeed & Ali Saeed · Qiskit Fall Fest 2026 · 354 spoken words ≈ 2:45 at a calm pace

BEFORE YOU RECORD

- Open Report.pdf and switch to full-screen or presentation view. The page numbers below are the PDF page numbers.
- In a terminal in the project folder, run python run_experiments.py --quick (about 30 s) so the output is ready to show.
- Record with OBS or the Xbox Game Bar (Win+G) at 1080p. Keep it under 3:00, because judges stop watching at the limit.
- Split the lines between Usman and Ali if you like: for example, one reads segments 1–3 and the other reads 4–7.

THE SCRIPT

SAY = speak these words *SHOW = what is on screen while you say them*

0:00 – 0:15 1. HOOK Report page 1

SAY

"Hi, we're Team Winners: Usman Saeed and Ali Saeed. Our project is fair load-shedding with quantum optimisation."

SHOW

Report page 1 (cover). Open Report.pdf full-screen on this page.



p.1

0:15 – 0:40 2. THE PROBLEM Report page 4

SAY

"Every summer, Pakistani distribution companies like LESCO and K-Electric face a supply gap and must decide, hour by hour, which feeders go dark. Shed too little and the grid trips. Shed too much and paying customers suffer for nothing. And a hospital must never lose power."

SHOW

Page 4, The Challenge. Point at the "Our test grid" sidebar on the right while you mention the hospital.



p.4

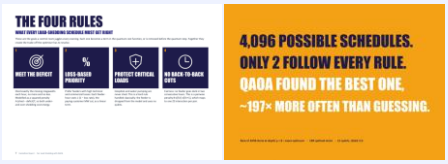
0:40 – 1:05 3. FOUR RULES, TWELVE QUBITS Report pages 7, 2

SAY

"We turned this into four rules: meet the megawatt deficit exactly, prefer high-loss feeders the way Pakistan's loss-based load management does, never cut critical loads, and never cut the same feeder two hours in a row. Our test case has four feeders over three evening hours. Each yes-or-no decision is one qubit: twelve qubits, 4,096 possible schedules, and only two follow every rule."

SHOW

Page 7, The Four Rules: point at each tile as you name it. At "twelve qubits" switch to page 2 (the amber statement page).



p.7 · p.2

1:05 – 1:35 4. HOW WE SOLVE IT Report pages 6, 9

SAY

"The hospital feeder is removed classically, so that rule can never break. The other rules become a QUBO cost function: the deficit and fairness rules are penalties, and fairness is a natural two-qubit interaction. We map it to an Ising Hamiltonian in Qiskit and run QAOA: alternating cost and mixer layers, tuned by a classical optimiser. We use CVaR-QAOA, which focuses on the best ten percent of outcomes."

SHOW

Page 6, From Grid to Qubits (the objective box in the middle). At "we map it to an Ising Hamiltonian" switch to page 9, the architecture diagram, and trace the hybrid loop with your cursor.



p.6 · p.9

1:35 – 2:15 5. DEMO AND RESULTS Report pages 11, 12, 13

SAY

"Here's our code running. The exact optimum sheds F2 and F4 at six, F1 at seven, and F2 and F3 at eight, and QAOA's best measured schedule is exactly that. The greedy rule a control room might use cuts forty megawatts at seven when only thirty were needed. As we add QAOA layers, the chance of measuring the optimum climbs to about five percent, roughly two hundred times better than guessing, and from depth six the most-measured schedule is the optimum."

SHOW

Start with the terminal: show the finished output of python run_experiments.py --quick (run it before you start recording). Then page 11 (schedules), page 12 (depth chart: point at the rising dark line), and finally page 13 (the tall amber bar).



p.11 · p.12 · p.13

2:15 – 2:45 6. HONEST LIMITATIONS Report page 14

SAY

“Now the honest part. A classical computer solves this in milliseconds, so there's no quantum speedup here. At fifteen qubits QAOA almost never found the optimum, because the penalties make the landscape rugged. And on real noisy hardware, a depth-eight circuit with over seven hundred two-qubit gates wouldn't survive.”

SHOW

Page 14, The Limits. Point at tiles 1, 8 and 5 as you mention them.



p.14

2:45 – 3:00 7. CLOSE Report pages 15, 16

SAY

“Next we'd try constraint-preserving mixers and split the problem by hour. Fair load-shedding is a real, daily problem that maps cleanly onto qubits. Our code is on GitHub. Thank you!”

SHOW

Page 15 (next steps) for the first sentence, then end on page 16, the Thank You page with the GitHub link.



p.15 · p.16

AFTER RECORDING Upload to Google Drive → Share → “Anyone with the link”. Paste the link into the report (cover and back page) and the submission form, then open it in a private window to check it plays.